cells. The membrane is absent in centriole (non-membranous organelle) (Figure 6.21).

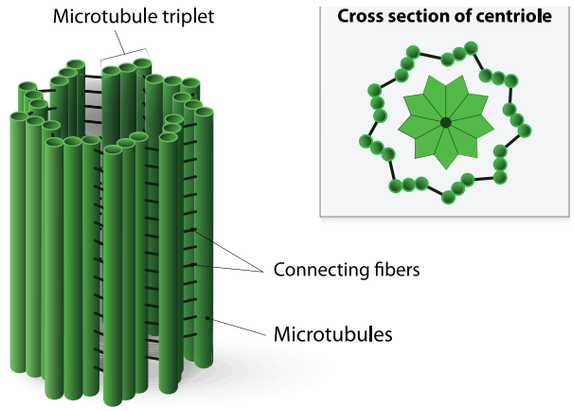


Figure 6.21: Structure of Centriole

6.6.14 Vacuoles

In plant cells vacuoles are large, bounded by a single unit membrane called **Tonoplast**. The Vacuoles contain cell sap, which is a solution of sugars, amino acids, mineral salts, waste chemical and anthocyanin pigments. Beetroot cells contain anthocyanin pigments in their vacuoles. Vacuoles accumulate products like tannins. The osmotic expansion of a cell kept in water is chiefly regulated by vacuole and the water enters the vacuole by osmosis.

The major function of plant vacuole is to maintain water pressure known as **turgor pressure**, which maintains the plant structure. Vacuoles organises itself into a storage/ sequestration compartment. Example: Vacuoles store, most of the sucrose of the cell.

i. Sugar in Sugar beet and Sugar cane.
ii. Malic acid in Apple.
iii. Acids in Citrus fruits.
iv. Flavonoid pigment cyanidin 3 rutinoside in the petals of *Antirrhinum*.

6.7. Nucleus

Nucleus is an important unit of cell which controls all activities of the cell. Nucleus holds the hereditary information. It is the largest among all cell organelles. It may be spherical, cuboidal, ellipsoidal or discoidal.

It is surrounded by a double membrane structure called **nuclear envelope**, which has the inner and outer membrane. The inner membrane is smooth without ribosomes and the outer membrane is rough by the presence of ribosomes and it continues with irregular and infrequent intervals with the endoplasmic reticulum. The membrane is perforated by pores known as **nuclear pores** which allows materials such as mRNA, ribosomal units, proteins and other macromolecules to pass in and out of the nucleus. The pores enclosed by circular structures called **annuli**. The pore and annuli form the **pore complex**. The space between two membranes is called **perinuclear space**.

DO YOU KNOW?

Chromatin is a viscous gelatinous substance that contains DNA, histone & non–histone proteins and RNA. H1, H2A, H2B, H3 and H4 are the different histones found in chromatin. It is formed by a series of repeated units called nucleosomes. Each nucleosome has a core of eight histone subunits.

Nuclear space is filled with **nucleoplasm**, a gelatinous matrix has uncondensed **chromatin** network and a conspicuous **nucleolius**. The Chromatin network is an uncoiled, indistinct and remain thread like during the interphase. It has little amount of RNA and DNA bound to histone proteins in eukaryotic cells (Figure 6.22).

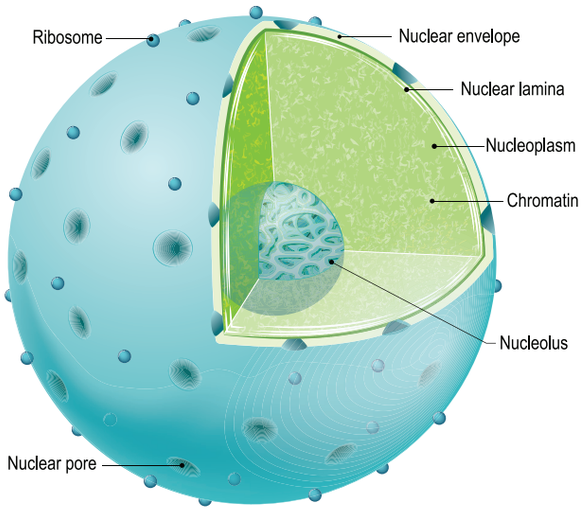


Figure 6.22: Structure of a Nucleus

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